

Proposal in Response to RFQ – Solar-Powered Irrigation Systems

Submitted to: Supremo Green Revolution (SGR)

Submitted by: AquaFuture International (AFI)

Date: October 6, 2025

1. Executive Summary

AquaFuture International (AFI) is pleased to present this proposal in response to the Supremo Green Revolution (SGR) Request for Quotation issued in September 2025 for the supply and installation of solar-powered irrigation systems across Kenya, Ghana, and Tanzania.

AFI is a global development organization specializing in sustainable water and energy solutions for rural communities. With over 12 years of project delivery experience in Africa and Asia, we combine engineering expertise with a strong focus on community empowerment and capacity building.

This proposal outlines a practical, cost-effective, and socially inclusive approach to delivering high-quality solar irrigation systems that align with SGR's mission to improve food security and farmer resilience. AFI's competitive advantage lies in its proven success implementing similar projects under low-resource conditions and its commitment to long-term sustainability through training, monitoring, and partnership models.

2. Organizational Profile

Founded in 2013, AquaFuture International operates from regional hubs in Nairobi, Accra, and Hyderabad. The organization employs engineers, agronomists, and social mobilization specialists who work together to design and deploy scalable renewable energy systems in rural areas.

AFI is ISO 9001:2015 certified and follows ISO 14001:2015 environmental management principles, though formal certification for ISO 45001 is currently in progress. AFI has successfully implemented over 50 renewable energy projects globally, including solar pumping, desalination, and community-scale water distribution.

Key strengths include:

- A multi-disciplinary team experienced in solar irrigation design and farmer training.
- Long-term partnerships with ministries of agriculture in East and West Africa.
- Established supplier network with local assembly and spare parts sourcing.
- Proven monitoring and evaluation framework based on UNDP and FAO standards.

3. Technical Approach

AFI proposes a modular solar-powered irrigation system suitable for smallholder plots between 0.5 and 5 hectares. Each unit includes a solar PV array (2–5 kW), DC submersible pump, water storage tanks (up to 5,000 liters), and a drip irrigation network optimized for high-efficiency water use.

Key design features:

- Pre-engineered modular kits configurable by site size and water depth.
- Anti-corrosion stainless steel frames with IP65-rated electronics.

- 10-year solar panel lifespan and 3-year pump warranty.
- GSM-enabled controllers for optional performance monitoring.
- Integration of smart irrigation timers to prevent overwatering.

Installation and commissioning will follow a 4-phase workflow:

1. **Baseline Survey & Site Selection (Month 1)**** – Assessment of solar resource, soil, and water conditions.
2. **System Design & Procurement (Month 2)**** – Finalization of specifications, procurement, and shipment.
3. **Installation & Farmer Training (Months 3–4)**** – Field installation, testing, and on-site hands-on training.
4. **Monitoring & Reporting (Month 5 onward)**** – Quarterly impact and performance reporting to SGR.

4. Deliverables

1. Design, supply, and installation of 15 solar irrigation systems across Kenya, Ghana, and Tanzania.
2. Training for 600 farmers and 30 local technicians in equipment operation, maintenance, and financial literacy.
3. Provision of pictorial training manuals in English, Swahili, and Twi.
4. Quarterly monitoring reports covering water output, uptime, and community impact.
5. Final completion and sustainability report with recommendations for scale-up.

5. Compliance and Standards

AFI ensures compliance with the following standards:

- ISO 9001:2015 – Quality Management Systems
- ISO 14001:2015 – Environmental Management Systems
- IEC 62257 – Renewable Energy Off-grid Guidelines
- SDG 6, 7, and 13 Alignment – Water, Clean Energy, and Climate Action Goals

In addition, AFI adheres to local labor and environmental laws in all three project countries and conducts annual internal audits to ensure continuous improvement.

6. Financial Proposal (Indicative, USD)

Item	Quantity	Unit Cost (USD)	Subtotal (USD)
Solar Irrigation System (complete package)	15	6,000	90,000
Farmer Training Sessions	15	2,000	30,000
Spare Parts and Maintenance Kits	15	400	6,000
Monitoring & Evaluation	1	12,000	12,000
Project Management and Logistics	1	15,000	15,000
Total (Excl. Taxes & Freight)			**153,000**

Notes:

- Prices exclude VAT, customs clearance, and inland logistics.
- Payment milestones: 40% upon contract signing, 40% on installation completion, 20% after final acceptance.
- All figures valid for 90 days from submission.

7. Sustainability and After-Sales Plan

AFI's sustainability model focuses on local ownership and skill transfer. Key elements include:

- Creation of community-based maintenance groups trained to handle first-line repairs.
- Partnerships with microfinance institutions to enable cost recovery and reinvestment.
- Two-year after-sales service contract covering preventive maintenance and spare part supply.
- Integration with national agricultural extension services for ongoing support.
- Use of recyclable materials and minimal packaging to reduce environmental footprint.

8. Strengths and Risks

****Strengths:****

- Strong training and social mobilization framework.
- Proven implementation record with donor-funded projects.
- Moderate cost structure with transparent pricing.
- Commitment to ISO 9001/14001 compliance.

****Risks / Weaknesses:****

- Limited in-house engineering capacity for high-complexity sites.
- Partial certification for ISO 45001 (Health & Safety) still pending.
- Heavy reliance on subcontractors for logistics in Ghana and Tanzania.

9. Conclusion

AquaFuture International is confident that its combination of technical competence, participatory approach, and focus on sustainability aligns closely with SGR's goals. While AFI acknowledges the need to strengthen its on-ground technical workforce, its commitment to training, community engagement, and performance transparency ensures long-term value for SGR and its partner farmers.

We look forward to the opportunity to collaborate with Supremo Green Revolution (SGR) in delivering renewable, inclusive agricultural innovation across Africa.

Submitted by:

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